Fuel-Optimal Trajectory Design and Mode-Switching Analysis for Asteroid Exploration Using Dual-Mode Propulsion

Haiyi Wang
National University of Defense Technology, Changsha, 410073, China

Xiang Guo
National University of Defense Technology, Changsha, 410073, China

Zhen Yang
National University of Defense Technology, Changsha, 410073, China

Yazhong Luo
National University of Defense Technology, Changsha, 410073, China

***Abstract*—This paper investigates fuel-optimal trajectory design and mode-switching characteristics for asteroid transfer missions using a single-thruster dual-mode propulsion system. The propulsion system contains a low-thrust high-specific-impulse mode and a high-thrust low-specific-impulse mode, which respectively provide propellant efficiency and enhanced maneuvering capability. Based on modified equinoctial elements and indirect optimal control theory, a fuel-optimal dual-mode trajectory optimization model is formulated. The optimal thrust direction, switching functions, Hamiltonian-based mode competition rule, and two-point boundary-value problem are derived. A hyperbolic-tangent smoothing homotopy strategy is introduced to improve the convergence of the bang-bang fuel-optimal control problem. An Earth-to-Ceres heliocentric transfer is used as the numerical case. Four propulsion schemes, namely low-thrust-only, high-thrust-only, dual-mode with coasting, and dual-mode without coasting, are compared in terms of terminal mass and switching behavior. In addition, a two-dimensional parameter plane defined by the thrust ratio and specific-impulse ratio is used to identify the mass-gain boundary of the dual-mode scheme relative to the low-thrust baseline. Numerical results show that the dual-mode coasting scheme can provide fuel benefits under specific transfer-time constraints and propulsion-parameter combinations. The high-thrust low-specific-impulse mode is mainly activated over short critical arcs for trajectory correction, whereas the low-thrust high-specific-impulse mode dominates long-duration propulsion. As the transfer time increases, the high-thrust mode fraction decreases and the terminal mass increases. The parametric boundary analysis indicates that the dual-mode advantage is jointly governed by thrust enhancement and specific-impulse loss, with the specific-impulse ratio showing a stronger influence on mass gain. These results provide useful guidance for mode design, propulsion-parameter matching, and mission analysis of dual-mode deep-space transfers.**

***Index Terms*—Dual-mode propulsion, Fuel-optimal trajectory, Indirect method, Mode switching, Asteroid transfer, Continuous-thrust Trajectory optimization**

Manuscript received XXXXX 00, 0000; revised XXXXX 00, 0000; accepted XXXXX 00, 0000. Corresponding author: Xiang Guo.

Haiyi Wang, PhD Student, College of Aerospace Science and Engineering; also State Key Laboratory of Space System Operation and Control, Changsha 410073, China (wanghaiyi@nudt.edu.cn).
Xiang Guo, Lecturer, College of Aerospace Science and Engineering; also State Key Laboratory of Space System Operation and Control, Changsha 410073, China (gx9645@126.com).
Zhen Yang, Professor, College of Aerospace Science and Engineering; also State Key Laboratory of Space System Operation and Control, Changsha 410073, China (yangzhen@nudt.edu.cn).
Yazhong Luo, Professor, College of Aerospace Science and Engineering; also State Key Laboratory of Space System Operation and Control, Changsha 410073, China (luoyz@nudt.edu.cn).

This work was supported in part by the U.S. Department of Commerce under Grant

I. Introduction

Deep-space exploration has become an important frontier in space science, resource utilization, and aerospace capability development. Current deep-space missions are gradually evolving from single flyby and imaging missions to long-duration residence, complex rendezvous, sample return, and multi-target exploration [1]. Accordingly, trajectory optimization has become a key basis for mission concept assessment, propulsion-system selection, and engineering implementation [2]. Among various deep-space targets, asteroids and dwarf planets have both fundamental scientific value and potential engineering significance. Asteroids preserve information on the material composition and dynamical evolution of the early Solar System, and they are also relevant to planetary defense, in-situ resource utilization, and future deep-space infrastructure [3].

For heliocentric deep-space missions such as small-body rendezvous, sample return, and main-belt exploration, trajectory design is usually affected by long flight duration, large accumulated velocity increment, mission-window sensitivity, and strict propellant limitations. Rayman et al. [15] showed that solar electric propulsion can support long-duration main-belt asteroid exploration missions. Dachwald et al. [16] further demonstrated that continuous-thrust propulsion can maintain high propellant efficiency over a relatively wide mission window in high-velocity-increment scenarios such as sample return. Therefore, long-duration heliocentric transfer missions are not only typical applications of continuous-thrust trajectory optimization, but also important problems for assessing the value of new propulsion-mode designs.

For these missions, continuous-thrust propulsion has become an important technical route for deep-space mission design because of its high specific impulse, high propellant utilization, and suitability for long-duration accumulation of velocity increment. Jiang et al. [6] presented practical homotopic indirect techniques for fuel-optimal low-thrust problems, and Chi et al. [7] further investigated time-optimal and fuel-optimal indirect optimization with a gridded ion thruster model. Russell [8] carried out global low-thrust trade studies based on primer vector theory. The surveys by Morante et al. [17], Betts [18], Conway [19], and Shirazi et al. [20] also indicate that low-thrust trajectory design has developed into a relatively systematic theoretical and numerical methodology.

In continuous-thrust trajectory optimization, the indirect method can directly provide the optimal thrust direction, switching structure, and transversality conditions, which gives it distinctive advantages for high-precision trajectory design. However, its numerical convergence strongly depends on the selection of state variables, the construction of initial costates, and the smoothing strategy for the control law. The modified equinoctial elements proposed by Walker et al. [21] provide an important basis for low-thrust trajectory modeling, and Junkins and Taheri [22] discussed the applicability of different state-vector choices in low-thrust optimization. The direct-to-indirect mapping of Ottesen and Russell [23], the quadratic homotopy method of Pan et al. [24], the enhanced smoothing method of Taheri et al. [25], and the generic smoothing method proposed by Taheri and Junkins [26] all show that high-quality initial-guess construction and nonsmooth-control treatment remain key difficulties in engineering applications of indirect methods. Taheri et al. [27] further extended propulsion systems with multiple operating modes to a more general multimode trajectory-design framework, providing a methodological basis for subsequent studies on dual-mode propulsion indirect methods.

In recent years, with the increasing demand for discrete propulsion operating points, power constraints, and trajectory-propulsion co-design, the research scope of low-thrust trajectory optimization has gradually expanded from classical fixed-performance propulsion to more general multiple-operating-mode problems. Arya et al. [28], [29] performed coupled optimization of trajectory design and propulsion-system parameters, and applied an optimal multimode propulsion model to low-thrust gravity-assist missions. Chi et al. [30], Li et al. [31], and Woollands et al. [32] studied power-limited trajectories, dual-homotopy solution strategies, and perturbed low-thrust transfers, respectively. Yam et al. [33], Englander and Conway [34], Sun et al. [35], and Kluever [36] improved the engineering usability of low-thrust mission design from the perspectives of global optimization, automated solution, and preliminary design. Mengali and Quarta [37], Quarta and Mengali [38], and Nah et al. [39] further enriched the understanding of tradeoffs and optimal-solution characteristics for two-body and heliocentric deep-space low-thrust transfers.

Although continuous-thrust propulsion can significantly reduce propellant consumption, a pure low-thrust scheme often requires a long transfer time, whereas a pure high-thrust scheme has difficulty maintaining propellant efficiency. Therefore, propulsion concepts combining a high-thrust low-specific-impulse mode and a low-thrust high-specific-impulse mode have attracted increasing attention. The studies by Kluever [40], [42], Oleson et al. [43], Mailhe and Heister [44], and Oh et al. [45] showed that chemical-electric combined propulsion can provide a more flexible tradeoff between propellant consumption and transfer time in orbit raising, GEO transfers, and general deep-space missions. However, most of these studies remain at the mission-level comparison stage, and the optimal mode-switching solution under shared-propellant conditions is still limited.

Compared with conventional hybrid propulsion, a key feature of multimode propulsion is that different operating modes share propellant and the propulsion-system platform. Rovey et al. [9] systematically reviewed the concepts and categories of multimode propulsion from the propulsion-system perspective. Their work clarifies the distinction between multimode propulsion and general hybrid propulsion, and points out that such systems can significantly expand mission capability boundaries. The Composite Smooth Control framework proposed by Taheri et al. [10], [27] provides a unified smoothing-based optimal-control modeling approach for propulsion systems with multiple operating modes. Its advantage is that discrete mode switching can be embedded into differentiable continuous controls, which facilitates indirect-method solution. Nevertheless, it is more oriented toward general multimode modeling, and the mutually exclusive operation constraint and specific mode-selection law of a single thruster are still not fully characterized.

Cline et al. [11] first established an indirect minimum-fuel and constrained-minimum-time trajectory optimization model with automatic mode-selection capability for a shared-monopropellant dual-mode spacecraft. They presented independent switching functions, smoothed throttles, mode Hamiltonian functions, and mutually exclusive mode-selection rules, which form the most directly relevant theoretical basis for the present study. Subsequently, Cline et al. [12] analyzed propellant benefits, switch counts, and minimum switching times for lunar SmallSat missions with dual-mode propulsion. Huang et al. [13] applied the idea of independent switching, global mode selection, and homotopic continuation to the fuel-optimal low-thrust problem of multimode propulsion, further verifying the engineering feasibility of this class of methods. Falcone et al. [14] compared single, hybrid, and multimode propulsion architectures from the perspective of system capability envelopes, while Herman et al. [46], Quarta [47], Gagne et al. [48], Berg and Rovey [49], Koizumi et al. [50], Mani et al. [51], Quarta et al. [52], and Alnaqbi et al. [53] demonstrated the potential value of combined or multimode propulsion in near-Earth asteroid, Mars, CubeSat, and small-satellite missions. However, existing studies are mainly focused on mission-feasibility demonstrations or specific scenario validations. Systematic comparisons of different on/off switching schemes, mode-switching mechanisms, and propulsion-parameter matching rules for long-duration heliocentric deep-space transfers remain insufficient.

A further review of the existing literature reveals three issues that deserve further investigation for dual-mode propulsion deep-space transfers. First, existing studies focus more on whether a dual-mode optimal solution can be obtained, whereas systematic comparisons among low-thrust-only, high-thrust-only, dual-mode with coasting, and dual-mode without coasting schemes are still insufficient. Second, most existing results remain at the level of numerical optimal trajectories, and there is a lack of dedicated discussion, in the context of deep-space transfers, on when mode switching occurs, whether coasting arcs are necessary, and which mode is dominant during different mission phases. Third, for dual-mode propulsion-parameter design, especially whether there exists a clear critical relationship between the thrust enhancement and the specific-impulse loss of the high-thrust mode, existing work still lacks quantitative criteria that can be used for mission assessment and propulsion-parameter matching.

In this paper, an Earth-to-Ceres heliocentric fuel-optimal transfer is selected as the numerical case, and a single-thruster dual-mode indirect-method model is established using modified equinoctial elements. The propulsion system contains a low-thrust high-specific-impulse mode and a high-thrust low-specific-impulse mode. On the basis of the automatic mode-selection idea of Cline et al. [11], four schemes are formulated in a unified manner: low-thrust-only, high-thrust-only, dual-mode with coasting, and dual-mode without coasting. The propellant consumption and mode-switching behavior of these schemes are compared. Furthermore, a two-dimensional parameter plane is constructed using the thrust ratio and specific-impulse ratio of the high-thrust mode relative to the low-thrust mode. By continuously varying the thrust and specific-impulse combinations of the two propulsion modes, the mass-gain distribution and critical boundary of the dual-mode scheme relative to the low-thrust-only scheme are established, providing references for propulsion-parameter matching, mode design, and mission analysis of dual-mode deep-space exploration missions.

II. Methodology

A. Dynamical Model

This paper considers the Earth-to-Ceres fuel-optimal rendezvous problem in a heliocentric two-body dynamical environment, and uses modified equinoctial elements to describe the spacecraft orbital state. To avoid the singularities of classical orbital elements near zero eccentricity and zero inclination, and to write the dynamics in a control-affine form under continuous thrust, the modified equinoctial elements are adopted as the state variables and are defined as

$$\begin{cases} p = a(1-e^2), \\ f = e\cos(\Omega+\omega), \\ g = e\sin(\Omega+\omega), \\ h = \tan\left(\dfrac{i}{2}\right)\cos\Omega, \\ k = \tan\left(\dfrac{i}{2}\right)\sin\Omega, \\ L = \Omega+\omega+\nu \end{cases} \tag{1}$$

where a, e, i, Ω, ω, and ν denote the semimajor axis, eccentricity, inclination, right ascension of the ascending node, argument of periapsis, and true anomaly, respectively. Since the Ceres transfer is a long-duration heliocentric flight and the thrust acceleration is small relative to the central gravitational acceleration, the modified equinoctial-element dynamics can be written in a unified form consisting of a natural drift term and a control-affine term. This form facilitates the subsequent construction of the Hamiltonian and the derivation of the switching functions [6], [7].

Let the spacecraft state vector be $\boldsymbol{x} = \begin{bmatrix} p & f & g & h & k & L \end{bmatrix}^{\mathrm{T}}$, and let the mass state be m. For a single-thruster dual-mode system, mode 1 is defined as the low-thrust high-specific-impulse mode and mode 2 as the high-thrust low-specific-impulse mode. The two modes share the same thrust-direction vector $\boldsymbol{\alpha} = \begin{bmatrix} \alpha_r & \alpha_t & \alpha_n \end{bmatrix}^{\mathrm{T}}$, satisfying $\|\boldsymbol{\alpha}\| = 1$. At any time, the normalized throttle of mode i is denoted by δ_i, with $0 \le \delta_i \le 1$. In the dual-mode propulsion system, the two modes share the same propellant and nozzle; only the thrust magnitude and specific impulse are changed through operating-point switching. Therefore, the total thrust acceleration can be written as $\boldsymbol{a}_T = (T_1\delta_1 + T_2\delta_2)\boldsymbol{\alpha}/m$, and the mass flow rate can be expressed as a linear superposition of the mass flow rates of the two modes. Since at most one propulsion mode is allowed to operate at any time, the following condition is imposed:

$$\delta_1, \delta_2 \in [0,1], \qquad \delta_1\delta_2 = 0 \tag{2}$$

The total thrust acceleration is decomposed into radial, transverse, and normal components in the orbital frame and substituted into Gauss' variational equations written in modified equinoctial elements. The effect of the thrust input on the element rates can then be arranged as the product of the matrix $\boldsymbol{M}(\boldsymbol{x})$ and the direction vector $\boldsymbol{\alpha}$, while the natural evolution of the elements caused only by heliocentric two-body gravity is included in the drift term [6], [7]. The system state equations can therefore be written as

$$\begin{aligned} \dot{\boldsymbol{x}} &= \frac{T_1\delta_1 + T_2\delta_2}{m}\boldsymbol{M}(\boldsymbol{x})\boldsymbol{\alpha} + \boldsymbol{D}(\boldsymbol{x}), \\ \dot{m} &= -\frac{T_1}{c_1}\delta_1 - \frac{T_2}{c_2}\delta_2 \end{aligned} \tag{3}$$

where

$$\boldsymbol{M}(\boldsymbol{x})=\sqrt{\frac{p}{\mu}}\begin{bmatrix}0 & 2p/w & 0\\ \sin L & [(w+1)\cos L+f]/w & -g(h\sin L-k\cos L)/w\\ -\cos L & [(w+1)\sin L+g]/w & f(h\sin L-k\cos L)/w\\ 0 & 0 & s^2\cos L/(2w)\\ 0 & 0 & s^2\sin L/(2w)\\ 0 & 0 & (h\sin L-k\cos L)/w\end{bmatrix}\tag{4}$$

$$\boldsymbol{D}(\boldsymbol{x})=\begin{bmatrix}0 & 0 & 0 & 0 & 0 & \sqrt{\frac{\mu}{p^3}}w^2\end{bmatrix}^{\mathrm{T}},\tag{5}$$

$$\begin{aligned}w&=1+f\cos L+g\sin L,\\ s^2&=1+h^2+k^2,\\ c_i&=I_{\mathrm{sp},i}g_0\end{aligned}\tag{6}$$

where T_i and $I_{\mathrm{sp},i}$ denote the thrust magnitude and specific impulse of mode i, respectively, and g_0 is the standard gravitational acceleration. The vector $\boldsymbol{D}(\boldsymbol{x})$ represents the natural drift of the modified equinoctial elements under solar gravity alone, and the matrix $\boldsymbol{M}(\boldsymbol{x})$ represents the influence of radial, transverse, and normal thrust accelerations on the element rates. In essence, $\boldsymbol{M}(\boldsymbol{x})$ is a 6×3 thrust-influence matrix assembled from the coefficients of Gauss' variational equations, with each column corresponding to the contribution of radial, transverse, and normal thrust components to the element rates. The term $\boldsymbol{D}(\boldsymbol{x})$ represents the natural drift of the modified equinoctial elements under thrust-free motion, in which only the true longitude L has an explicit drift term, while the other elements remain constant under the pure two-body assumption.

The present study considers a heliocentric fuel-optimal rendezvous problem with a given departure time and transfer time. Let the modified equinoctial elements of Earth and Ceres at the corresponding epochs be $\boldsymbol{x}_{\mathrm{E}}(t_0)$ and $\boldsymbol{x}_{\mathrm{C}}(t_{\mathrm{f}})$, respectively. The boundary conditions can then be written as

$$\boldsymbol{x}(t_0)=\boldsymbol{x}_{\mathrm{E}}(t_0),\qquad m(t_0)=m_0,\qquad \boldsymbol{x}(t_{\mathrm{f}})=\boldsymbol{x}_{\mathrm{C}}(t_{\mathrm{f}}),\tag{7}$$

$$t_{\mathrm{f}}=t_0+T_{\mathrm{tr}}\tag{8}$$

Because this paper focuses on comparing the propellant cost of different propulsion modes under the same mission window, the transfer time is not treated as an optimization variable. Instead, it is prescribed as a given parameter for the fair comparison of the four schemes considered below [7], [11].

B. Fuel-Optimal Indirect Method for Dual-Mode Propulsion

To compare propellant consumption under different propulsion modes in a unified manner, this paper adopts an integral fuel-optimal performance index, i.e., the propellant mass consumed by the two modes over the whole transfer is minimized directly. For the dual-mode propulsion system, the instantaneous propellant mass flow rates of the two modes are proportional to T_1/c_1 and T_2/c_2, respectively. Therefore, the objective functional cannot be simply written as a throttle integral; the thrust and specific-impulse parameters of different modes must be retained explicitly. The corresponding objective functional is shown below, and its physical meaning is to directly minimize the total propellant consumption over the entire flight [11].

$$J=\int_{t_0}^{t_{\mathrm{f}}}\left(\frac{T_1}{c_1}\delta_1+\frac{T_2}{c_2}\delta_2\right)dt\tag{9}$$

Introduce the state costate $\boldsymbol{\lambda}_x$ and the mass costate λ_m to construct the Hamiltonian. To apply Pontryagin's minimum principle, the state costate and mass costate $\boldsymbol{\lambda}_x=\begin{bmatrix}\lambda_p & \lambda_f & \lambda_g & \lambda_h & \lambda_k & \lambda_L\end{bmatrix}^{\mathrm{T}}$ and λ_m are defined.

Substituting the state equations and the mass equation yields the Hamiltonian

$$\begin{aligned}H=&\boldsymbol{\lambda}_x^{\mathrm{T}}\left[\frac{T_1\delta_1+T_2\delta_2}{m}\boldsymbol{M}(\boldsymbol{x})\boldsymbol{\alpha}+\boldsymbol{D}(\boldsymbol{x})\right]\\ &+(1-\lambda_m)\left(\frac{T_1}{c_1}\delta_1+\frac{T_2}{c_2}\delta_2\right)\end{aligned}\tag{10}$$

This expression combines the propellant-consumption term, the orbital-state dynamics, and the mass dynamics into a single scalar function, providing a unified starting point for subsequently deriving the thrust direction, mode throttles, and costate equations.

The term in the Hamiltonian associated with the thrust direction $\boldsymbol{\alpha}$ can be written as

$$H_\alpha=\frac{T_1\delta_1+T_2\delta_2}{m}\left(\boldsymbol{M}^{\mathrm{T}}\boldsymbol{\lambda}_x\right)^{\mathrm{T}}\boldsymbol{\alpha}\tag{11}$$

Because $\boldsymbol{\alpha}$ is a unit vector, for a given throttle and state costate, the optimization problem is equivalent to minimizing $\left(\boldsymbol{M}^{\mathrm{T}}\boldsymbol{\lambda}_x\right)^{\mathrm{T}}\boldsymbol{\alpha}$ subject to the constraint $\|\boldsymbol{\alpha}\|=1$. According to the Cauchy-Schwarz inequality,

$$\left(\boldsymbol{M}^{\mathrm{T}}\boldsymbol{\lambda}_x\right)^{\mathrm{T}}\boldsymbol{\alpha}\geq-\left\|\boldsymbol{M}^{\mathrm{T}}\boldsymbol{\lambda}_x\right\|,\tag{12}$$

and equality holds when $\boldsymbol{\alpha}$ and $\boldsymbol{M}^{\mathrm{T}}\boldsymbol{\lambda}_x$ are oppositely collinear. Therefore, the optimal thrust direction satisfies

$$\boldsymbol{\alpha}^*=-\frac{\boldsymbol{M}^{\mathrm{T}}(\boldsymbol{x})\boldsymbol{\lambda}_x}{\|\boldsymbol{M}^{\mathrm{T}}(\boldsymbol{x})\boldsymbol{\lambda}_x\|}\tag{13}$$

This result is consistent with the primer-vector opposite-direction law in the single-mode continuous-thrust fuel-optimal problem, indicating that the additional difficulty introduced by dual-mode propulsion mainly lies in mode selection rather than in the solution of the thrust direction [7], [10], [11].

After substituting the optimal direction into the Hamiltonian, the switching functions corresponding to the two modes can be obtained. After this substitution, the direction variable is eliminated, and the terms in the Hamiltonian associated with δ_i can be rearranged as

$$H=\boldsymbol{\lambda}_x^{\mathrm{T}}\boldsymbol{D}(\boldsymbol{x})+\delta_1S_1+\delta_2S_2,\tag{14}$$

where S_i is the switching-function coefficient of mode i. It characterizes the marginal influence of activating the mode on the Hamiltonian. With this arrangement, one obtains

$$S_i=\frac{T_i}{c_i}\left(\frac{c_i}{m}\left\|\boldsymbol{M}^{\mathrm{T}}\boldsymbol{\lambda}_x\right\|+\lambda_m-1\right),\qquad i=1,2\tag{15}$$

The switching function therefore contains the thrust magnitude, specific impulse, current mass, and primer-vector norm.

When strict bang-off-bang control is used, the switching function makes the right-hand side of the dynamics discontinuous near switching points, which introduces significant initial-guess sensitivity into the shooting solution. To improve numerical convergence, this paper adopts a hyperbolic-tangent smoothing homotopy and writes the throttle of each mode as

$$\delta_i = \frac{1}{2}\left[1 + \tanh\left(\frac{S_i}{\varepsilon}\right)\right], \qquad i = 1, 2, \tag{16}$$

$$\delta_i^* = \begin{cases} 1, & S_i > 0, \\ 0, & S_i < 0, \end{cases} \qquad i = 1, 2, \tag{17}$$

where ε is the homotopy parameter. During the solution process, ε is gradually continued from a relatively large smoothing value to a value close to zero to approximate the original nonsmooth fuel-optimal control. When ε is relatively large, the control function is continuous and easy to solve; as ε is gradually decreased, the smoothed control approaches the ideal bang-off-bang control.

For a dual-mode system sharing a single thruster, only one mode can operate at any time. Therefore, this paper constructs the mode Hamiltonian functions H_1 and H_2 corresponding to mode 1 and mode 2, and determines the actually activated mode by comparing their values. It should be noted that the preceding equations only describe the independent activation tendency of the two modes under their respective switching functions. For a shared single-thruster dual-mode system, an additional mutually exclusive mode-selection criterion is required. The mode Hamiltonians are constructed for the two cases in which mode 1 is active and mode 2 is off, and mode 2 is active and mode 1 is off, respectively. Comparing these two values determines which operating mode should be activated at the current time.

$$\delta_1^* = \begin{cases} \delta_1, & H_1 \le H_2, \\ 0, & H_1 > H_2, \end{cases} \tag{18}$$

$$\delta_2^* = \begin{cases} \delta_2, & H_2 < H_1, \\ 0, & H_2 \ge H_1, \end{cases} \tag{19}$$

where

$$H_1 = \boldsymbol{\lambda}_x^{\mathrm{T}}\left[\frac{T_1\delta_1}{m}\boldsymbol{M}(\boldsymbol{x})\boldsymbol{\alpha}^* + \boldsymbol{D}(\boldsymbol{x})\right] + (1 - \lambda_m)\frac{T_1}{c_1}\delta_1, \tag{20}$$

$$H_2 = \boldsymbol{\lambda}_x^{\mathrm{T}}\left[\frac{T_2\delta_2}{m}\boldsymbol{M}(\boldsymbol{x})\boldsymbol{\alpha}^* + \boldsymbol{D}(\boldsymbol{x})\right] + (1 - \lambda_m)\frac{T_2}{c_2}\delta_2 \tag{21}$$

In the above equations, when $S_1 < 0$ and $S_2 < 0$, one has $\delta_1 \approx 0$ and $\delta_2 \approx 0$, so the system naturally enters a coasting arc. The coasting dual-mode control law contains two levels of selection. First, S_1 and S_2 determine whether the two modes are worth activating; second, the comparison between H_1 and H_2 determines which feasible mode should be truly activated. When at least one switching function is positive, the candidate modes are selected according to the Hamiltonian minimum principle. This structure preserves the thrust-coast switching commonly seen in fuel-optimal problems and can also describe the internal mode-switching law of a dual-mode system [11], [13].

After the optimal direction and smoothed throttle expressions are obtained, the costate equations can be written directly as

$$\begin{aligned} \dot{\boldsymbol{\lambda}}_x &= -\frac{\partial H}{\partial \boldsymbol{x}}, \\ \dot{\lambda}_m &= -\frac{\partial H}{\partial m} = -\frac{T_1\delta_1^* + T_2\delta_2^*}{m^2}\left\|\boldsymbol{M}^{\mathrm{T}}\boldsymbol{\lambda}_x\right\| \end{aligned} \tag{22}$$

For the state costate $\boldsymbol{\lambda}_x$, its derivative is determined by the drift term, the matrix $\boldsymbol{D}(\boldsymbol{x})$, the matrix $\boldsymbol{M}(\boldsymbol{x})$, and their partial derivatives with respect to the state. For the mass costate λ_m, since the mass enters the Hamiltonian only through the thrust-acceleration term, it is sufficient to differentiate the $1/m$ term to obtain the explicit expression. This expression indicates that the evolution of λ_m is directly related to the total thrust level and the primer-vector norm, and therefore plays a key role in the mode-switching criterion [7], [8], [11].

Since both the initial and terminal states are prescribed, whereas the terminal mass is free, the boundary conditions must also satisfy the transversality condition associated with the free terminal mass, in addition to the state endpoint constraints given above:

$$\lambda_m(t_{\mathrm{f}}) = 0 \tag{23}$$

C. Two-Point Boundary-Value Problem

Section 2.2 transforms the dual-mode propulsion fuel-optimal problem into a two-point boundary-value problem with mode-selection logic. For numerical solution, under given initial state, initial mass, and transfer time, the initial costates satisfying the terminal constraints can be obtained by a single-shooting method.

With fixed departure epoch, fixed arrival epoch, and fixed initial mass, the initial orbital state and the terminal target orbital state are prescribed, whereas the terminal mass is free. Therefore, the single-shooting method in this paper takes the initial costates as the unknown variables. Let the shooting variables be

$$\boldsymbol{z} = \begin{bmatrix}\boldsymbol{\lambda}_x^{\mathrm{T}}(t_0) & \lambda_m(t_0)\end{bmatrix}^{\mathrm{T}} \tag{24}$$

Given this set of initial costates, the state equations, mass equation, costate equations, and optimal control law can be integrated from the initial time to the terminal time. According to the terminal orbital-state constraints and the transversality condition associated with the free terminal mass, the shooting equations for the fuel-optimal rendezvous problem can be constructed as

$$\boldsymbol{\Phi}_{F,\mathrm{rend}}\left(\boldsymbol{\lambda}_x(t_0); \lambda_m(t_0)\right) = \begin{bmatrix}\boldsymbol{x}(t_{\mathrm{f}}; \boldsymbol{z}) - \boldsymbol{x}_{\mathrm{f}}(t_{\mathrm{f}}) \\ \lambda_m(t_{\mathrm{f}}; \boldsymbol{z})\end{bmatrix} = \boldsymbol{0} \tag{25}$$

where $\boldsymbol{x}_{\mathrm{f}}(t_{\mathrm{f}})$ denotes the modified equinoctial elements of the target body at the arrival epoch. Solving this equation yields the initial costates that satisfy the state endpoint constraints and the terminal transversality condition, thereby completing the formulation of the two-point boundary-value problem for the fuel-optimal dual-mode propulsion problem.

D. Four Propulsion Schemes and Two-Dimensional Critical Boundary Definition

To systematically analyze the mechanism of dual-mode propulsion in the Earth-to-Ceres fuel-optimal transfer, the following four propulsion schemes are considered. Within the unified optimal-control framework, the differences among the four schemes do not appear in the state equations or boundary conditions, but in the admissible set of control variables. In other words, the four schemes can be regarded as special cases of the same fuel-optimal problem under different control constraints. By comparing these special cases under the same initial mass, boundary conditions, and transfer time, the effects of the number of modes, the availability of coasting, and the matching of mode parameters on propellant consumption can be identified more clearly.

(1) Low-thrust-only scheme: only mode 1 is retained, with $\delta_2 \equiv 0$, and the switching function of mode 1 determines the thrust-coast switching. This scheme corresponds to a high-specific-impulse low-thrust single-mode continuous-thrust fuel-optimal transfer. It is essentially a degeneration of the dual-mode model to a conventional single-mode indirect-method problem, and it serves as the baseline for evaluating whether dual-mode propulsion truly provides a fuel benefit.

$$\text{LT-only}: \quad \delta_2 = 0, \quad 0 \le \delta_1 \le 1 \tag{26}$$

(2) High-thrust-only scheme: only mode 2 is retained, with $\delta_1 \equiv 0$, and the switching function of mode 2 determines the thrust-coast switching. This scheme represents the propellant cost of a low-specific-impulse high-thrust single-mode propulsion system under the same transfer-time constraint, and it can be used to identify the extreme boundary at which propellant is exchanged for orbital maneuvering capability.

$$\text{HT-only}: \quad \delta_1 = 0, \quad 0 \le \delta_2 \le 1 \tag{27}$$

(3) Dual-mode with coasting: both modes are allowed to participate in the optimization, and the actual throttle is determined by the preceding equations. At most one mode can operate at any time, and both modes are allowed to be turned off simultaneously. Thus, the trajectory may contain switching between mode 1 and mode 2, as well as coasting arcs. This scheme is closest to the dual-mode propulsion control structure with coasting given by Cline et al. [11], and it is the main object used in this paper to investigate the coexistence of mode switching and coasting.

$$\text{Dual-coast}: \quad \delta_1^* + \delta_2^* \le 1 \tag{28}$$

(4) Dual-mode without coasting: both modes are allowed to participate in the optimization, and the actual throttle is determined by the corresponding equations. Because exactly one mode must operate at any time, the independent-switching-plus-coasting logic can no longer be used directly. Instead, a continuous competition relation must be constructed between the two modes so that the following condition is always satisfied:

$$\delta_1^* + \delta_2^* = 1 \tag{29}$$

Then the switching function is

$$\delta_1^* = \frac{1}{2}\left[1 + \tanh\left(\frac{S_1 - S_2}{\varepsilon_s}\right)\right], \qquad \delta_2^* = 1 - \delta_1^* \tag{30}$$

where ε_s is the mode-switching smoothing parameter.

Unlike the coasting dual-mode scheme, the above expression enforces $\delta_1^* + \delta_2^* = 1$, eliminating all coasting arcs and retaining only switching between mode 1 and mode 2. A mode-competition function is established by comparing the relative magnitudes of S_1 and S_2: when $S_1 - S_2 < 0$, mode 1 is more favorable; otherwise, mode 2 is more favorable. As ε_s is gradually reduced, the expression degenerates into an approximately hard switching control. Therefore, differentiability can be maintained during the solution process while the true mutually exclusive mode-switching law is approached.

This scheme is used to examine the influence of internal switching between the two modes on propellant consumption when coasting freedom is unavailable. It also provides a comparison with the coasting scheme to evaluate the value of coasting freedom itself.

$$\text{Dual-no-coast}: \quad \delta_1^* + \delta_2^* = 1 \tag{31}$$

To further study the applicability range of dual-mode propulsion relative to the low-thrust-only scheme, this paper describes the coupled influence of the two modes using a two-dimensional parameter plane formed by the thrust ratio and the specific-impulse ratio. Considering mode 1 as the low-thrust high-specific-impulse mode and mode 2 as the high-thrust low-specific-impulse mode, the thrust ratio and specific-impulse ratio of the high-thrust mode relative to the low-thrust mode are defined as

$$r_{\mathrm{T}} = \frac{T_2}{T_1}, \qquad r_{\mathrm{I}} = \frac{I_{\mathrm{sp},2}}{I_{\mathrm{sp},1}} \tag{32}$$

where the thrust and specific impulse of mode 1 are denoted by T_1 and $I_{\mathrm{sp},1}$, respectively, and those of mode 2 are denoted by T_2 and $I_{\mathrm{sp},2}$, respectively. Because this paper focuses on a single thruster with two thrust modes, the thrust ratio is generally greater than 1 and the specific-impulse ratio lies between 0 and 1. That is, mode 2 obtains higher thrust at the expense of specific impulse. The subsequent parameter scan therefore uses the two-dimensional combination of thrust ratio and specific-impulse ratio as the basic variables.

Under the same initial mass, orbital boundary conditions, transfer time, and number of revolutions, let the terminal mass obtained by the dual-mode coasting scheme be the dual-mode terminal mass, and let the terminal mass obtained by the low-thrust-only scheme be the low-thrust-only terminal mass. This paper defines the mass gain of dual-mode propulsion relative to the low-thrust-only scheme as the difference in terminal mass:

$$\Delta m_{\mathrm{f}}(r_{\mathrm{T}}, r_{\mathrm{I}}) = m_{\mathrm{f}}^{\mathrm{DM}}(r_{\mathrm{T}}, r_{\mathrm{I}}) - m_{\mathrm{f}}^{\mathrm{LT}} \tag{33}$$

When the mass gain is greater than zero, introducing mode 2 and coasting freedom yields a net fuel benefit for the corresponding thrust-ratio and specific-impulse-ratio combination. When the mass gain is equal to zero, the fuel performance of the dual-mode scheme is equivalent to that of the low-thrust-only scheme. When the mass gain is less than zero, the propellant penalty caused by the high-thrust low-specific-impulse mode exceeds its orbital maneuvering benefit for that parameter combination. Because the admissible control set of the dual-mode coasting scheme contains the low-thrust-only operating mode, its global optimum should theoretically not be inferior to the low-thrust-only scheme.

Accordingly, the dual-mode advantage region and the equivalent boundary can be defined as

$$\Omega_+ = \{(r_{\mathrm{T}}, r_{\mathrm{I}}) \mid \Delta m_{\mathrm{f}}(r_{\mathrm{T}}, r_{\mathrm{I}}) > 0\}, \tag{34}$$

$$\Omega_0 = \{(r_{\mathrm{T}}, r_{\mathrm{I}}) \mid \Delta m_{\mathrm{f}}(r_{\mathrm{T}}, r_{\mathrm{I}}) = 0\}, \tag{35}$$

where the equivalent boundary is the two-dimensional critical boundary used in this paper. Considering convergence errors in practical indirect-method solutions and discretization errors in parameter sampling, a small mass-gain threshold may be used in subsequent simulation plots to approximate the zero-gain boundary, namely

$$B_{\varepsilon} = \{(r_{\mathrm{T}}, r_{\mathrm{I}}) \mid \Delta m_{\mathrm{f}}(r_{\mathrm{T}}, r_{\mathrm{I}}) = b\}, \tag{36}$$

where the mass-gain threshold b can be selected as a small positive value such as 0.1 kg or 1 kg to improve the identifiability of the boundary line in heat maps. By grid-based calculation of the fuel-optimal solutions of the dual-mode coasting scheme in the thrust-ratio-specific-impulse-ratio plane and comparison with the terminal mass of the low-thrust-only baseline solution, the mass gain, zero-gain boundary, and critical curves under different gain thresholds can be obtained. This two-dimensional critical boundary can simultaneously reflect the coupled influence of thrust enhancement and specific-impulse reduction on the effectiveness of dual-mode propulsion. It is therefore suitable for describing mode-switching behavior and fuel-benefit distributions of dual-mode propulsion.

III. Simulation Results and Analysis

To verify the applicability of the dual-mode propulsion indirect method to fuel-optimal asteroid-transfer design, the simulation study is organized at three levels. First, typical transfer trajectories under different transfer-time parameters are used to examine the correspondence between automatic mode selection and orbital-element evolution. Second, under the same mission background, the fuel performance of four schemes, namely low-thrust-only, high-thrust-only, dual-mode with coasting, and dual-mode without coasting, is compared. Third, in the two-dimensional parameter plane formed by the thrust ratio and the specific-impulse ratio, the advantage boundary of the dual-mode scheme relative to the low-thrust-only baseline is characterized. This organization makes it possible to observe the control structure inside a single optimal trajectory and to provide design criteria for propulsion-parameter selection.

A. Typical Dual-Mode Transfer Trajectories and Mode-Switching Characteristics

The time parameter for typical trajectories is denoted by α_t, which maps different transfer times into a unified dimensionless interval. In this study, the all-high-thrust time-optimal result is used as the lower bound and is defined as t^*_{HT}, with $t^*_{\mathrm{HT}} = 444.766466$ d; the all-low-thrust time-optimal result is used as the upper bound and is defined as t^*_{LT}, with $t^*_{\mathrm{LT}} = 1358.190915$ d. For a prescribed α_t, the transfer time t_{tr} is determined by linear interpolation $t_{\mathrm{tr}} = t^*_{\mathrm{HT}} + \alpha_t(t^*_{\mathrm{LT}} - t^*_{\mathrm{HT}})$, equivalently $\alpha_t = (t_{\mathrm{tr}} - t^*_{\mathrm{HT}})/(t^*_{\mathrm{LT}} - t^*_{\mathrm{HT}})$. Thus, a larger α_t corresponds to a longer allowed transfer time. Table II lists three typical dual-mode coasting cases. When α_t increases from 0.60 to 0.90, the transfer time increases from 992.8 d to 1266.8 d, and the terminal mass increases from 784.651 kg to 1209.719 kg. This trend shows that, as the time margin increases, the optimal solution can allocate more velocity increment to long low-thrust high-specific-impulse arcs and complete part of the phase adjustment through natural drift. The high-thrust low-specific-impulse mode is activated only over local arcs where strong orbital reconstruction is required. The corresponding heliocentric transfer trajectories are shown in Fig. 1.

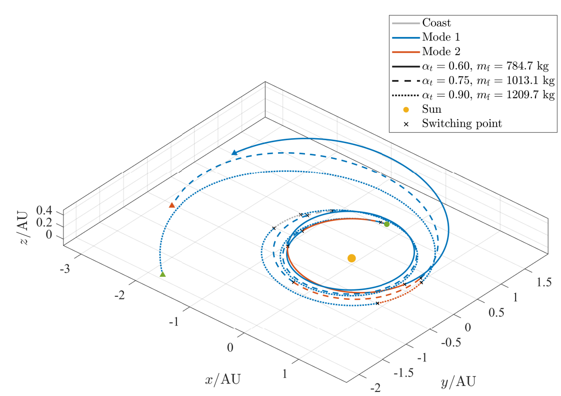


Fig. 1. Dual-mode coasting transfer trajectories under different α_t.

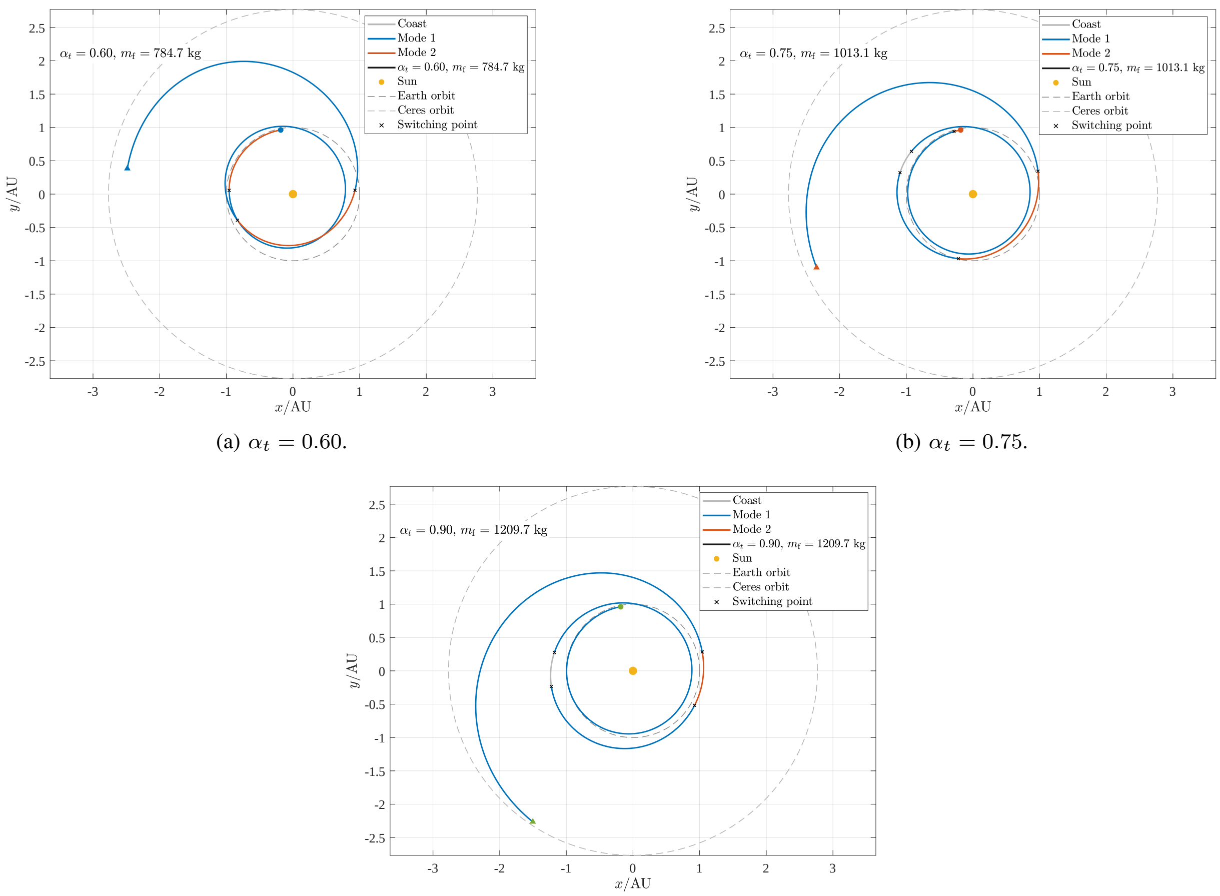
The corresponding planar projections in the heliocentric x-y plane are further shown in Fig. 2, where the local high-thrust arcs and switching points can be observed more clearly for each transfer-time parameter.

All three transfer trajectories show outward expansion from a near-Earth heliocentric orbit to an orbit near Ceres, but the expansion rates are different. For a smaller α_t, the orbital-energy increase is more concentrated and the

TABLE I

Basic parameters and boundary conditions of the simulation case

Parameter	Value
Initial mass m_0	2000 kg
T_1, $I_{\mathrm{sp},1}$	0.235 N, 4155 s
T_2, $I_{\mathrm{sp},2}$	0.5 N, 1000 s
g_0	$9.80665\ \mathrm{m/s^2}$
Heliocentric state of Earth	$\boldsymbol{r}_E(t_0) = (-2.758794880 \times 10^7,\ 1.439239583 \times 10^8,\ 1.921064327 \times 10^4)$ km; $\boldsymbol{v}_E(t_0) = (-29.77686365,\ -5.53581334,\ -1.94338794 \times 10^{-4})$ km/s
Heliocentric state of Ceres	$\boldsymbol{r}_C(t_0) = (3.248137504 \times 10^8,\ -2.963900580 \times 10^8,\ -6.932647919 \times 10^7)$ km; $\boldsymbol{v}_C(t_0) = (11.19565321,\ 12.13914566,\ -1.676178255)$ km/s
Terminal boundary	The terminal modified equinoctial elements of the spacecraft are equal to the target state of Ceres propagated to t_{f}.



(a) $\alpha_t = 0.60$.

(b) $\alpha_t = 0.75$.

(c) $\alpha_t = 0.90$.

Fig. 2. Heliocentric x-y projections of typical dual-mode coasting trajectories under different α_t.

TABLE II

Parameters of typical dual-mode coasting cases

α_t	Transfer time/d	Revolutions	Terminal mass/kg
0.60	992.8	2	784.651
0.75	1129.8	2	1013.061
0.90	1266.8	2	1209.719

mode-switching points appear earlier in key outward-migration arcs; as α_t increases, the expansion becomes smoother and the low-specific-impulse high-thrust mode is activated only over local short arcs. This indicates that dual-mode optimal control does not distribute thrust between the two modes according to a fixed ratio. Instead, the switching functions determined by the costates call the high-thrust mode only in local intervals where the demands for orbital energy, angular momentum, and plane adjustment are strongest.

The mode-switching histories further reveal the bang-off-bang structure of the dual-mode coasting solution. The three typical trajectories are all dominated by the high-specific-impulse low-thrust mode, and the high-thrust mode appears only in a few short arcs. For the cases with

$\alpha_t = 0.75$ and $\alpha_t = 0.90$, clear coasting arcs also appear, indicating that continuous thrust over the entire transfer is not necessarily optimal in the fixed-time fuel-optimal problem. When natural drift can complete phase waiting or avoid unnecessary orbital-element perturbations, coasting is directly converted into propellant savings, as shown in more detail in the next subsection.

The mode-time fractions provide a more direct fuel-mechanism interpretation. At $\alpha_t = 0.60$, the high-thrust-mode fraction is about 18.56%, indicating that a short transfer requires stronger rapid maneuvering capability. When α_t increases to 0.90, the high-thrust-mode fraction decreases to 3.58%, while the low-thrust-mode fraction increases to 93.61%. This trend is consistent with the increase in terminal mass, showing that the value of the high-thrust mode is mainly to relax time constraints and support local orbital reconstruction, rather than to replace the high-specific-impulse mode in long-duration propulsion.

The time histories of the orbital elements explain the dynamical causes of mode switching. In Figs. 5a and 5b, the light-blue background denotes intervals in which mode 1, the low-thrust high-specific-impulse mode, is active, the light-orange background denotes intervals in which mode 2, the high-thrust low-specific-impulse mode, is active, and unshaded narrow bands denote coasting arcs. The high-thrust background usually corresponds to intervals of rapid semimajor-axis increase or eccentricity adjustment, whereas the coasting arcs mostly appear when the orbital elements vary slowly or when phase matching is required. The inclination changes gradually over the transfer, indicating that the trajectory requires not only energy increase but also continuous plane correction. The high-thrust mode can accelerate local element variations, but because of its low-specific-impulse penalty, the optimal solution does not use it for long durations. The low-thrust high-specific-impulse mode therefore undertakes most of the energy accumulation and plane-adjustment task.

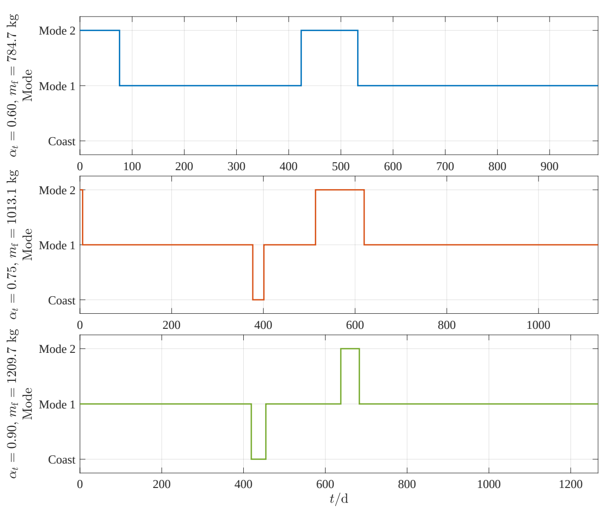


Fig. 3. Mode-switching histories of typical cases.

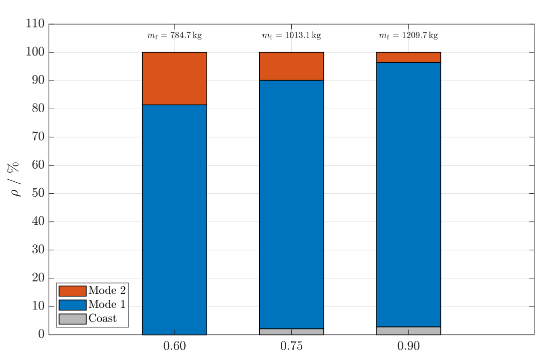


Fig. 4. Mode-time fractions and terminal mass of typical cases.

B. Fuel-Performance Comparison of the Four Propulsion Schemes

The comparison of the four schemes is not only used to compare terminal mass, but also to separate the effects of propulsion operating point and control freedom on fuel performance. The low-thrust-only and high-thrust-only schemes provide performance boundaries for single operating points, whereas the dual-mode coasting and dual-mode noncoasting schemes further examine the effects of mutually exclusive mode selection and coasting freedom. The terminal mass as α_t varies is shown in Fig. 6, and representative data are listed in Table III. Fig. 6 shows that the high-thrust-only curve remains at the lowest level, the dual-mode coasting curve is close to or slightly higher than the low-thrust-only curve in the medium-to-long time region, and the dual-mode noncoasting curve drops sharply when $\alpha_t > 1$. Although the high-thrust-only scheme has stronger acceleration capability, its low specific impulse makes its overall fuel performance much worse than schemes dominated by the low-thrust high-specific-impulse mode. Its terminal mass remains low, indicating that high thrust without sufficient specific impulse is difficult to translate into fuel advantage in long-duration deep-space transfers. The low-thrust-only scheme can obtain a high terminal mass when enough time is available, and in the interval $\alpha_t = 1.00$–1.20 it nearly overlaps the dual-mode coasting scheme. This means that within this time range, the high-specific-impulse low-thrust mode can already complete the main transfer task independently, although its feasibility and rapid maneuvering capability remain limited under tighter time constraints.

The dual-mode coasting scheme nearly overlaps the low-thrust-only scheme in the interval $\alpha_t = 1.00$–1.20. At $\alpha_t = 1.00$, its terminal-mass gain relative to the low-thrust-only scheme is about 1.822 kg, whereas at $\alpha_t = 1.10$ and $\alpha_t = 1.20$ the terminal masses of the two schemes listed in Table III are identical. This overlap indicates that when the transfer time approaches or exceeds the all-low-thrust time-optimal baseline, mode 1 can independently complete the main energy accu-

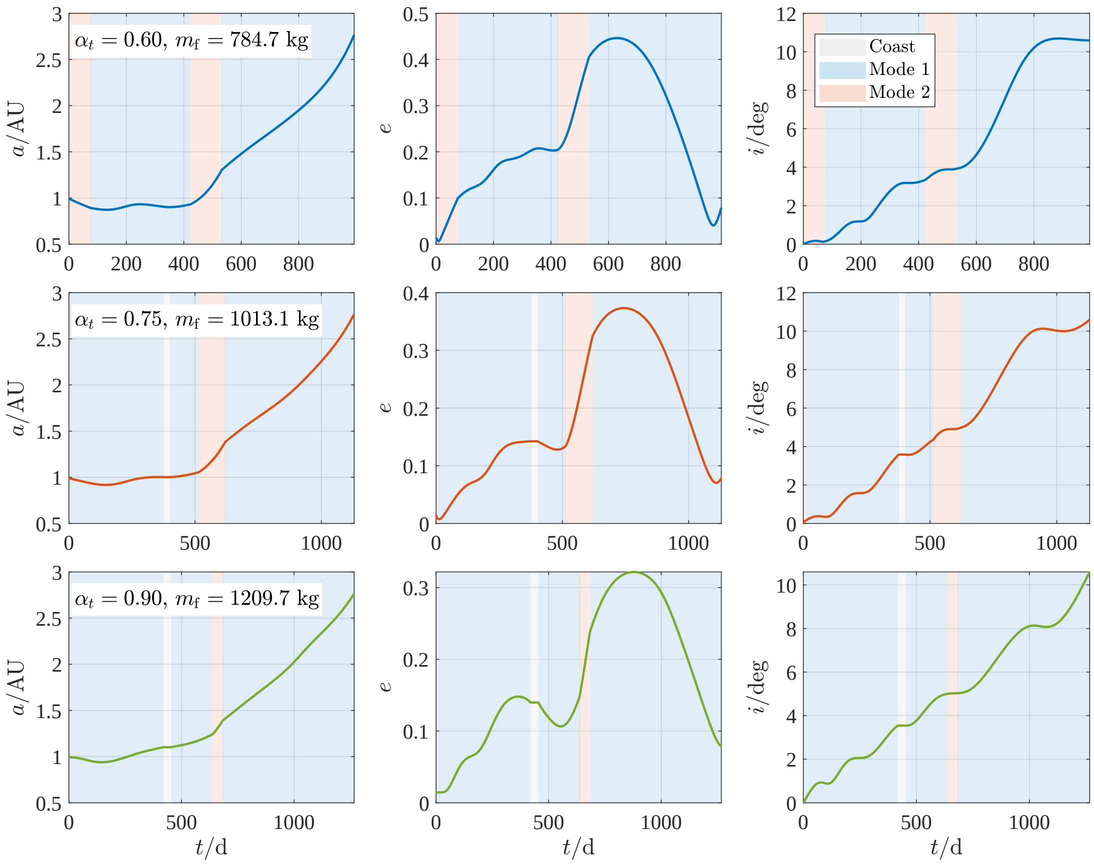


Fig. 5a. Variations of semimajor axis, eccentricity, and inclination with mode background.

TABLE III
Representative terminal masses of the four schemes, kg (optimal revolution number in parentheses)

α_t	Low-thrust only	High-thrust only	Dual-mode with coasting	Dual-mode without coasting
0.60	–	550.4(1)	1098.3(1)	1040.4(1)
0.75	–	553.7(1)	1100.5(1)	1012.6(2)
0.90	–	556.1(1)	1209.7(2)	1207.9(2)
1.00	1323.2(2)	574.2(2)	1325.0(2)	1323.2(2)
1.10	1413.7(2)	591.9(2)	1413.7(2)	919.5(1)
1.20	1451.6(2)	591.9(2)	1451.6(2)	904.7(1)

mulation and phase adjustment. The high-thrust low-specific-impulse mode no longer provides an obvious net benefit, and the dual-mode coasting optimum automatically degenerates into a solution dominated by the low-thrust mode. In contrast, the dual-mode noncoasting curve exhibits a distinct jump after $\alpha_t > 1$ because this scheme enforces thrust over the entire transfer and cannot use the extra time for natural drift and phase waiting. Table III also shows that its optimal revolution number switches from two revolutions to one revolution, changing the control structure and orbital winding pattern. Therefore, the terminal mass decreases discontinuously, indicating that coasting freedom is decisive for the fuel-optimal solution in long time windows.

The influence of coasting freedom on the fixed-time fuel-optimal problem is particularly significant. At $\alpha_t = 1.20$, the terminal mass of the dual-mode coasting scheme is 1451.641 kg, whereas that of the dual-mode noncoasting scheme is only 904.694 kg. Although the latter also has two thrust modes, it consumes unnecessary propellant in long-duration transfer because thrust must be applied throughout the trajectory. This result shows that the advantage of dual-mode propulsion comes not only from adding one more mode, but also from allowing the optimal control to freely choose among thrusting, coasting, and different modes.

The variation of mode fractions explains the performance differences above. The dual-mode coasting scheme exhibits clear coasting arcs under medium and larger α_t, whereas the dual-mode noncoasting scheme can only switch between the two propulsion modes and cannot use natural drift to save propellant. When a longer transfer time is allowed, forced continuous thrust destroys the fuel advantage of phase-waiting intervals. When the mission

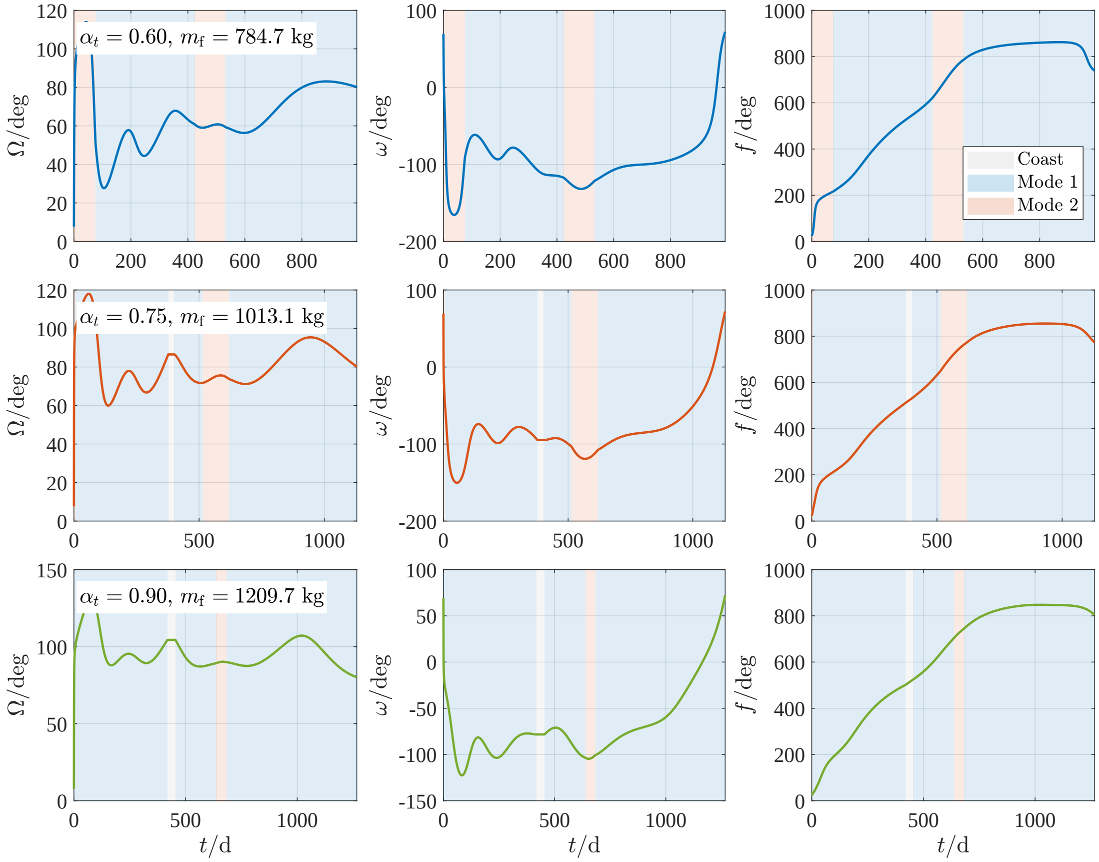


Fig. 5b. Variations of RAAN, argument of periapsis, and true anomaly with mode background.

time is tight, the high-thrust mode provides the necessary rapid maneuvering capability. Therefore, the thrust/coast sequence in dual-mode trajectory design should not be prescribed in advance. It should be automatically determined by switching functions and mode Hamiltonians, and dual-mode propulsion should not be interpreted simply as continuous alternation between two modes.

C. Two-Dimensional Critical Boundary Analysis

Under the baseline dual-mode parameters, the fuel advantage of the dual-mode coasting scheme relative to the low-thrust-only scheme is small. It is therefore necessary to further investigate the influence of propulsion-parameter matching on the advantage boundary. This paper uses the low-thrust-only result with $\alpha_t = 1.00$ and two revolutions as the baseline, whose terminal mass is 1323.215004 kg. With the low-thrust-mode parameters

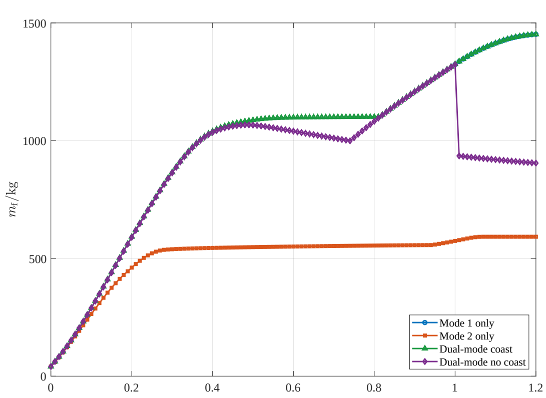


Fig. 6. Terminal mass of the four propulsion schemes as a function of α_t.

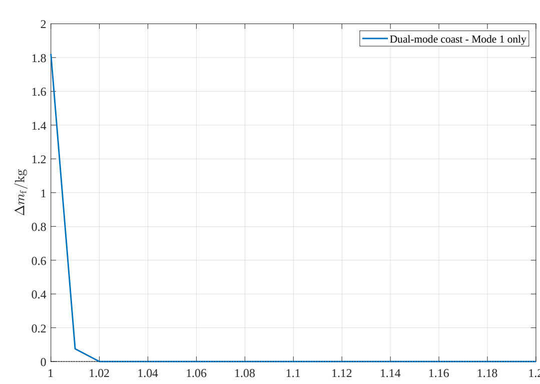


Fig. 7. Terminal-mass gain of the dual-mode coasting scheme relative to the low-thrust-only scheme.

fixed, the two-dimensional parameters $(r_{\mathrm{T}}, r_{\mathrm{I}})$ are varied, and the terminal-mass gain of the dual-mode coasting scheme is computed in the two-dimensional plane formed by the thrust ratio and the specific-impulse ratio. This representation avoids compressing the four propulsion parameters into a single composite index and is more suitable for observing the coupled relation between thrust enhancement and specific-impulse loss.

The continuously interpolated mass-gain Δm_{f} distribution shows that the lower-left region has zero mass gain, whereas the upper-right region contains positive gains. The dual-mode advantage boundary is inclined overall from the low-thrust-ratio, high-specific-impulse-ratio region toward the high-thrust-ratio, low-specific-impulse-ratio region. The physical meaning is that when the thrust advantage of the high-thrust mode is weak, its specific impulse must remain close to that of the low-thrust mode to offset the propellant penalty caused by low specific impulse. As the thrust ratio increases, the high-thrust mode can complete local orbital reconstruction in a shorter time, and the required specific-impulse ratio is therefore reduced. The maximum mass gain appears near a thrust ratio of 3.50 and a specific-impulse ratio of

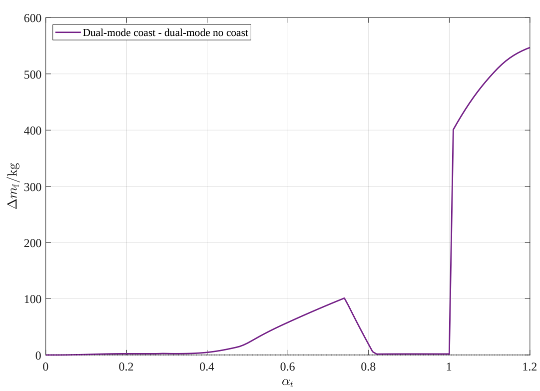


Fig. 8. Terminal-mass gain of the dual-mode coasting scheme relative to the dual-mode noncoasting scheme.

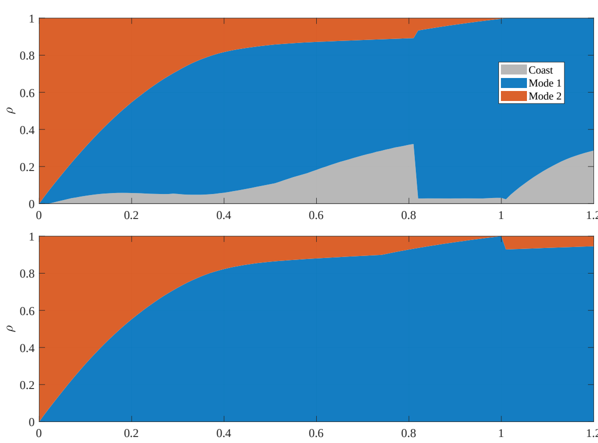


Fig. 9. Mode-time fractions of the dual-mode schemes as α_t varies.

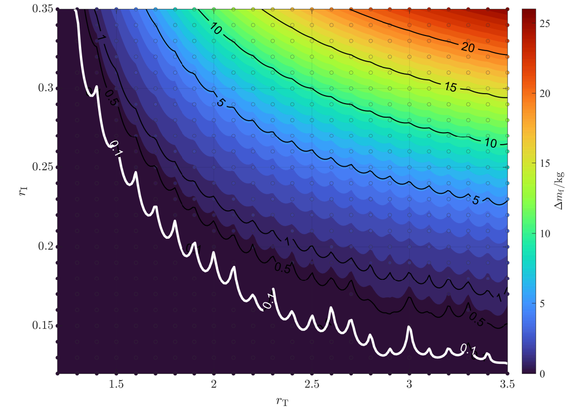


Fig. 10. Dual-mode mass gain and critical boundary in the thrust-ratio-specific-impulse-ratio plane.

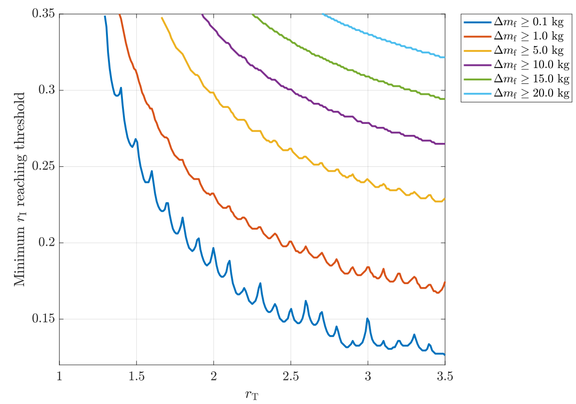


Fig. 11. Minimum specific-impulse-ratio curves corresponding to different mass-gain thresholds.

0.35, with a gain of about 25.619 kg and a corresponding terminal mass of 1348.834 kg.

The mass-gain-threshold B_{ε} curves further transform the two-dimensional results into propulsion-parameter selection criteria. For a small gain on the order of 1 kg, increasing the thrust ratio can clearly reduce the required specific-impulse ratio. For gains greater than 10 kg, the curves shrink noticeably toward the high-specific-impulse-ratio region, indicating that large benefits require not only high thrust capability but also a sufficiently high specific impulse in the high-thrust mode. The parameter region satisfying a mass gain no smaller than $\Delta m_{\mathrm{f}} = 1$ kg accounts for about 57.17%, the region satisfying no smaller than $\Delta m_{\mathrm{f}} = 5$ kg accounts for about 18.99%, and the region satisfying no smaller than $\Delta m_{\mathrm{f}} = 10$ kg accounts for only about 3.68%. Therefore, a high-benefit dual-mode scheme does not rely simply on high thrust, but on the combined satisfaction of high thrust and acceptable specific-impulse loss.

From the perspective of mode usage, the non-low-thrust mode fraction ρ and the mass gain Δm_{f} show a strong correlation, with a correlation coefficient of about 0.969. The correlation coefficient between the high-thrust-

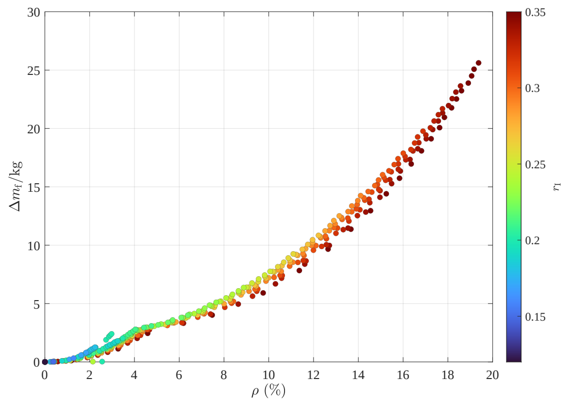


Fig. 12. Relationship between the non-low-thrust mode fraction ρ and mass gain Δm_{f}.

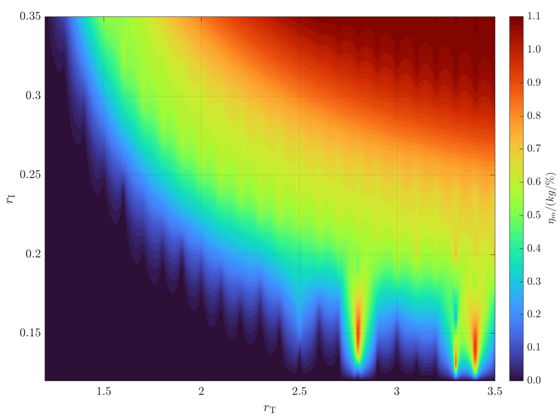


Fig. 13. Mass-gain efficiency per unit non-low-thrust mode fraction.

mode fraction and the mass gain is about 0.873. This indicates that the mass benefit of dual-mode propulsion does not come from simply extending the high-thrust operation time, but from the combination of short high-thrust maneuvers and necessary coasting. The point with the largest high-thrust-mode fraction is located near a thrust ratio of 2.10 and a specific-impulse ratio of 0.35, where the high-thrust-mode fraction is about 2.12%. However, this point is not the maximum mass-gain point. This phenomenon indicates that optimal mode selection must account for both orbital-dynamics benefit and propellant consumption, rather than monotonically increasing the use time of the high-thrust mode.

On the basis of the preceding correlation analysis, it is still necessary to distinguish the efficiency with which different parameter combinations use the non-low-thrust modes. To separate parameter regions that obtain mass gain by relying on relatively long non-low-thrust arcs from those that obtain effective benefits with a small non-low-thrust fraction, this paper combines the mass gain with the non-low-thrust mode fraction and defines the mass-gain efficiency per unit non-low-thrust mode fraction as

$$\eta_m = \frac{\Delta m_{\mathrm{f}}}{\rho}, \tag{37}$$

where

$$\rho = \frac{t_{\mathrm{c}} + t_2}{t_{\mathrm{tr}}}, \tag{38}$$

where η_m denotes the mass-gain efficiency per unit non-low-thrust mode fraction, ρ denotes the non-low-thrust mode-time fraction, and t_{c} and t_2 denote the coasting time and the high-thrust-mode operation time, respectively. The gain-efficiency plot provides another evaluation perspective: it measures how much terminal-mass benefit can be obtained per unit non-low-thrust mode fraction under limited high-thrust or coasting arcs. The high-efficiency region usually lies near the advantage boundary rather than in the upper-right region where the absolute gain is largest. This means that, in engineering selection, if the high-thrust mode is constrained by thermal control, power, or engine lifetime, parameter combinations near the boundary may be more attractive than the maximum-gain point because they obtain a clear fuel benefit with a lower high-thrust usage fraction.

Combining the two-dimensional gain distribution, threshold curves, and efficiency distribution shows that terminal mass generally increases with the thrust ratio and specific-impulse ratio, but the non-low-thrust mode fraction ρ is not fully synchronized with terminal mass. Some high-terminal-mass regions rely on relatively large coasting or high-thrust fractions, whereas some boundary regions obtain positive gain with limited use of non-low-thrust modes. This result has direct implications for single-thruster dual-mode design: if engineering constraints limit the operating duration of the high-thrust mode, mass-gain efficiency should be prioritized; if the mission emphasizes absolute propellant saving, parameter regions with both high thrust ratio and high specific-impulse ratio should be selected.

IV. Conclusions

This paper established and applied an indirect trajectory-optimization model for the Earth-to-Ceres fuel-optimal transfer using a single thruster with two thrust modes. Typical trajectories, scheme comparisons, and propulsion-parameter boundaries were systematically analyzed through numerical simulations. The results show that the optimal control structure of the dual-mode coasting trajectory has a clear bang-off-bang characteristic. The high-specific-impulse low-thrust mode undertakes the main long-duration propulsion task, the high-thrust low-specific-impulse mode is concentrated in local arcs for rapid reconstruction of orbital energy and shape, and coasting arcs are used for phase waiting and for avoiding unnecessary propellant consumption.

The comparison among the four schemes indicates that the advantage of dual-mode propulsion cannot be simply interpreted as having more operating modes. In a fixed-time fuel-optimal problem, coasting freedom is as important as mode switching. If the dual-mode system is forced to operate throughout the transfer, the additional mode may instead cause significant fuel loss during long-duration transfers. For the baseline propulsion parameters considered in this paper, the dual-mode coasting scheme has a clear advantage over the high-thrust-only scheme, whereas its benefit relative to the low-thrust-only scheme is small and concentrated in specific time windows. This indicates that whether dual-mode propulsion is worth adopting depends jointly on the mission time constraint and the matching of propulsion parameters.

The analysis in the thrust-ratio-specific-impulse-ratio plane provides a critical boundary with clearer engineering interpretation. Compared with a single composite critical parameter, the two-dimensional boundary directly reveals the compensation relation between thrust enhancement and specific-impulse penalty: increasing the thrust ratio can reduce the specific-impulse ratio required for positive gain, but large fuel benefits still require the high-thrust mode to maintain a sufficiently high specific impulse. The analyses of mode fraction and mass-gain efficiency further show that the optimal benefit comes from the synergy among short high-thrust maneuvers, coasting for phase waiting, and high-specific-impulse continuous propulsion, rather than from a monotonic increase in high-thrust-mode operation time.

The main value of this work is to transform the optimal-control results of dual-mode propulsion into analysis maps that can be used for mission design and preliminary propulsion-parameter selection. Although the theoretical framework is mainly based on existing indirect-method and dual-mode optimal-control techniques, the asteroid-transfer case reveals how different control freedoms, different α_t, and different propulsion-parameter combinations influence fuel performance. These results provide a relatively complete simulation basis for subsequent dual-mode deep-space trajectory optimization, parameter-sensitivity analysis, and English manuscript preparation. Future work may incorporate more refined ephemeris models, engine operating constraints, and homotopy solution strategies to improve convergence robustness near complex boundaries and extend the method to multi-target or multiple-small-body transfer missions.

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